Supplementary Material

Robust error-minimization in the genetic code across physicochemical metrics and variant codes: a graph-theoretic analysis in $GF(2)^6$

Paul Clayworth and Sergey Kornilov

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Roadmap. This supplement documents (i) the full claim hierarchy used to organise the manuscript's evidentiary tiers (§S1); (ii) sensitivity analyses for the central optimality and topology-avoidance results—encoding sweeps (§S2, §S4), the 2×2 definition $\times$ adjacency audit (§S3), candidate-universe denominators (§S5), conditional-logit IIA and clade-exclusion robustness (§S6, §S7), and a standard-code-proximity audit for the per-table claim (§S8); (iii) the topology-avoidance hypergeometric/permutation clade-exclusion analysis matched to Sengupta et al. (2007) (§S9); (iv) the complete tRNAscan-SE 2.0.12 dataset for the 18 verified genomes (§S10) and the de-duplicated reassignment database (§S11); (v) the synthetic-biology feasibility-score implementation referenced from main-text §2.5 (§S12); (vi) detailed KRAS–Fano falsification numbers (§S13); (vii) the full event-level conditional-logit specification including feature definitions, fitted coefficients (raw and normalised), likelihood-ratio tests, confounding diagnostics, and the order-averaging implementation (§S14); and (viii) software, version, and reproducibility metadata (§S15). All numbers reported here are rendered from the same `manuscript_stats.json` and per-analysis JSON artifacts that drive the main text, so the two documents cannot drift from each other within a single pipeline run.

S1 Claim hierarchy with full justifications

The 15 evaluated claims are organised into six evidentiary tiers (Supported / Suggestive / Exploratory / Falsified / Rejected / Tautological). The hierarchy is registered in code (`src/codon_topo/reports/claim_hierarchy.py` in https://github.com/biostochastics/codontopo) so that every analysis run produces a verifiable status report (`codon-topo claims`) rather than the status table being maintained as prose. The "Falsified" and "Rejected" rows record claims this paper or earlier work has tested and found to be wrong; we list them so that the framework's negative scope is visible alongside its positive results.

Claim ID	Status	Justification
hypercube_coloring_optimality	Supported	Cross-metric sensitivity: Grantham $p = 0.006$, Miyata $p < 0.001$, polar requirement $p = 0.003$, Kyte–Doolittle $p = 0.001$ (block-preserving null, n =10,000). Stop penalty sensitivity (0/150/215/300): immaterial.
per_table_optimality_preservation	Supported	Per-table block-preserving null applied to all 27 NCBI tables; significant fraction (BH–FDR $p < 0.05$) and per-table quantiles are reported in the main text and the per-table CSV `output/tables/T4_per_table_optimality.csv` (released in the `codontopo` repository). Translation table 3 (yeast mito) is the marginal exception.
optimality_rho_robustness	Supported	ρ-sweep at n =10,000: $p \leq 0.006$ at all ρ values; effect-size z increases monotonically from $\rho = 0$ to $\rho = 1$.
topology_avoidance_depletion	Supported	Permutation $p \leq 10^{-4}$ under both Q_6 and $H(3,4)$; hypergeometric $p = 1.6 \times 10^{-8}$ (Q_6, new disconnection) and $p = 1.3 \times 10^{-6}$ ($H(3,4)$, $\Delta\beta_0 > 0$). 5–7 of 28 de-duplicated events are topology-breaking versus $\approx$ 64–75% of the candidate landscape ($\approx$ 3.0–3.4-fold depletion). Q_6 is encoding-dependent (8 of 24 bijections give no depletion); $H(3,4)$ is encoding-independent and is the primary test. Clade-exclusion sensitivity (7 regimes): all $p < 10^{-5}$.
trna_enrichment_reassigned_aa	Suggestive	MIS worst-case $p = 0.045$. 18 tRNAscan-SE-verified assemblies (15 variant + 3 standard controls), 24 pairings across 5 variant codes.
bit_position_bias_weighted	Exploratory	Uniform $p = 0.006$ inflated by non-independence. De-duplicated $p = 0.075$.
mechanism_boundary_conditions	Exploratory	Three-tier pattern: duplication / stem shortening / modification. Descriptive.
atchley_f3_serine_convergence	Exploratory	Serine $F_3 = -4.760$, 2.24 SD below mean. Complementary, not independent.
variant_code_disconnection_catalogue	Exploratory	4 variant-code cases at $\varepsilon = 1$ in Q_6 (default encoding): Thr (Table 3, yeast mito), Leu (Tables 16/22, chlorophycean and *S. obliquus* mito; collapsed to one algal-mito event), Ala (Table 26, *Pachysolen*), Ser (Table 12, *Candida*). Separately, Table 32 (Balanophoraceae plastid, UAG→Trp) creates a 2-fold Trp pair (UGG, UAG) that remains connected at ε=1 but breaks the bit-5 two-fold filtration — a filtration finding, not a disconnection.
kras_fano_clinical_prediction	Falsified	$p = 1.0$ across all 6 G12 variants. n =1,670 MSK-IMPACT mutations.
serine_min_distance_4_invariant	Rejected	16/24 encodings give distance 2. Only 8/24 give distance 4.

Claim ID	Status	Justification
psl_2_7_symmetry	Rejected	No 64-dim irrep. Antoneli & Forger (2011).
holomorphic_embedding	Rejected	Domain is finite discrete. Character identity fails: $i^2 = -1 \neq 1$.
two_fold_bit_5_filtration	Tautological	Forced by encoding choice. Holds in 16/24 encodings.
four_fold_prefix_filtration	Tautological	Trivial under any bijection from 4 bases to $\mathrm{GF}(2)^2$.

Table 1: Complete claim hierarchy with justifications.

S2 Encoding sensitivity analysis (coloring optimality)

There are $4! = 24$ distinct bijections from $\{C, U, A, G\}$ to $\mathrm{GF}(2)^2$ (24 ways to assign the four nucleotide letters to the four 2-bit binary patterns 00/01/10/11). The default encoding used in the main text is $C \mapsto 00$, $U \mapsto 01$, $A \mapsto 10$, $G \mapsto 11$ (rationale stated in main-text §2.1). To check that the coloring-optimality result is not specific to that choice, we re-ran the cross-metric block-preserving null analysis under each of the 23 alternative encodings, holding everything else fixed. All 24 encodings yield significant optimality ($p < 0.05$ under the block-preserving null with $n =$10,000); the mean per-encoding quantile is 1.8%, confirming that the result is not an artifact of the default encoding.

Properties that are encoding-invariant:

- Serine disconnection at $\varepsilon = 1$ (holds under all 24 encodings)
- Coloring optimality significance (all 24 significant)
- Four-fold prefix filtration (tautological under any bijection)
- $H(3, 4)$ adjacency graph (encoding-independent by construction)

Properties that are encoding-dependent:

- Serine inter-family minimum Hamming distance (4 in 8/24 encodings, 2 in 16/24)
- Two-fold bit-5 filtration (holds in 16/24 encodings)
- Q_6 (Hamming-1) adjacency partition of $H(3, 4)$, and the topology-avoidance signal computed under Q_6 adjacency (see §S4)
- Specific score values and rank orderings

Full per-encoding results are reproducible from the `codontopo` repository via `codon-topo coloring --all-encodings`.

S3 Topology-breaking definitions: 2 × 2 audit

The topology-avoidance result depends on two independent definitional choices: (i) which adjacency graph defines codon-family connectivity (Q_6 Hamming-1 in the default $\mathrm{GF}(2)^6$ encoding, vs the encoding-independent $H(3, 4)$ single-nucleotide graph), and (ii) what counts as a "topology-breaking" candidate move (a candidate move creates a **new** disconnection in a previously connected family, vs the candidate move strictly increases β_0 summed across families). To prevent silent dependence on either choice, we report all four combinations.

We define two notions of "topology-breaking" candidate move:

1. **New disconnection in a previously connected family**: a candidate move makes some amino acid disconnected at $\varepsilon = 1$ that was connected in the standard code (i.e., the amino acid was not previously in the disconnection catalogue but is after the move).
2. **Increase in components** ($\Delta\beta_0 > 0$, the conditional-logit feature): the total number of connected components, summed across amino-acid codon graphs, strictly increases:

$$\Delta_{\text{topo}} = \sum_a \beta_0(G_a^{\text{after}}) - \sum_a \beta_0(G_a^{\text{before}}) > 0.$$

Both definitions are reported under both Q_6 adjacency (Hamming-1 in the default $\mathrm{GF}(2)^6$ encoding) and $H(3,4)$ adjacency (full single-nucleotide adjacency, encoding-independent), giving four cells. All four share the same denominators (1,280 candidate moves, 28 de-duplicated observed events).

Adjacency	Definition	K/N possible	x/n obs	Hyper. p	RR (95% CI)
$H(3,4)$ (primary)	$\Delta\beta_0 > 0$	846 / 1,280	6 / 28	1.3×10^{-6}	0.32 (0.16–0.66)
$H(3,4)$	new disconnection	822 / 1,280	5 / 28	5.0×10^{-7}	0.28 (0.13–0.62)
Q_6	$\Delta\beta_0 > 0$	963 / 1,280	7 / 28	2.2×10^{-8}	0.33 (0.17–0.63)
Q_6	new disconnection	931 / 1,280	6 / 28	1.6×10^{-8}	0.29 (0.14–0.60)

The four cells agree qualitatively: depletion is highly significant under all four combinations, with risk ratios in the range 0.28–0.33 and hypergeometric $p < 10^{-5}$ throughout. The main text uses the $H(3,4)$ Hamming graph with the **increase-in-components** ($\Delta\beta_0 > 0$) definition as the primary cell (encoding-independent adjacency, definition matched to the conditional-logit feature). The full machine-readable audit is released as part of the `codontopo` repository's `output/` artifacts (the `definitions_audit` block in the topology-avoidance results file).

S4 Topology avoidance: Q_6 encoding-sweep sensitivity

This section documents what motivated the demotion of Q_6 from primary to secondary topology adjacency in the manuscript.

The $H(3,4)$ Hamming graph is encoding-independent: every two-bit bijection from $\{A, C, G, U\}$ to $\{0,1\}^2$ produces the same nucleotide-level adjacency, because $H(3,4)$ depends only on which nucleotide letters differ between two codons, not on their binary encoding. The Q_6 subgraph, however, depends on the encoding because the partition into Hamming-1 (single-bit-change) edges versus Hamming-2 (within-nucleotide diagonal) edges is bijection-specific—under one encoding two codons that differ at exactly one nucleotide may project to a Hamming-1 edge of Q_6, under another encoding they may project to a Hamming-2 edge. To test whether the Q_6 topology-avoidance result survives this representation choice, we recomputed the Q_6 candidate-landscape rate, observed rate, depletion fold, and hypergeometric p-value under all 24 base-to-bit bijections, holding the same 1,280 candidate moves and 28 observed events.

Across all 24 encodings:

- candidate-landscape rate: min 36.0%, median 69.9%, max 72.7%
- observed rate: min 21.4%, median 28.6%, max 35.7%
- depletion fold: min 1.01, median 2.45, max 3.39
- hypergeometric p: min 1.6×10^{-8}, median 6.1×10^{-6}, max 0.572

The default encoding (C=00, U=01, A=10, G=11) gives the largest depletion and the smallest p. Eight of 24 encodings give a candidate-landscape rate of approximately 36%, under which the observed rate of 21–36% does not significantly differ from candidate ($p > 0.5$). The Q_6 result is therefore not encoding-invariant. We accordingly present the encoding-independent $H(3,4)$ result as the primary topology-avoidance test, with Q_6 reported as a representation-specific decomposition for continuity with the broader $\mathrm{GF}(2)^6$ framework. The default encoding was adopted in companion methodological work (Clayworth (2026)) prior to the present encoding sweep, on the grounds of visualization clarity (it places the standard code's nine 2-fold-degenerate amino acids on bit-5 differences). We make no claim that this encoding is biologically privileged. The full per-encoding sweep is released as part of the `codontopo` repository's `output/` artifacts (the `Q6_encoding_sweep` block in the topology-avoidance results file); Figure 1 visualizes the per-encoding depletion fold and hypergeometric p.

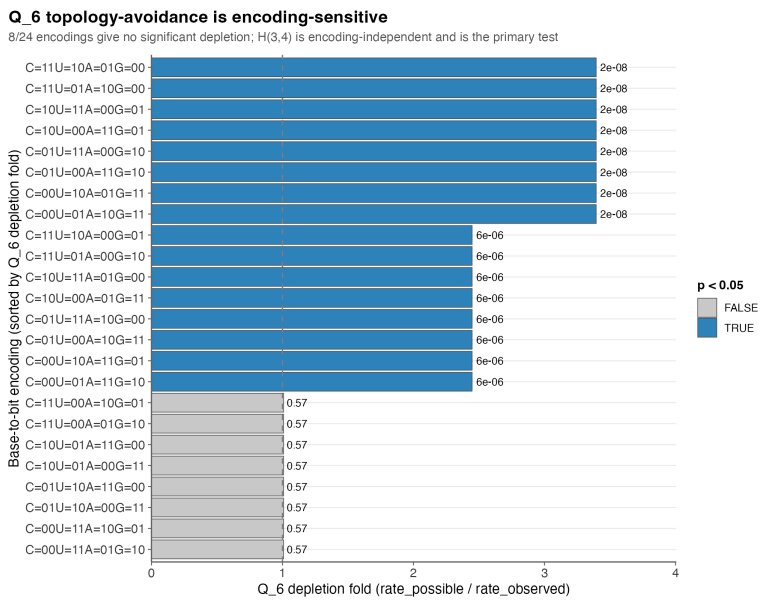

Figure 1: Per-encoding Q_6 topology-avoidance depletion across all 24 base-to-bit bijections. Bars are sorted by depletion fold; the dashed line at 1.0 marks no-depletion. Bars highlighted in blue are statistically significant ($p < 0.05$); grey bars are not. Eight of 24 encodings (depletion fold ≈ 1.0, $p > 0.5$) place the Q_6 candidate-landscape rate at $\approx 36\%$ rather than 73%, eliminating the depletion signal. The default encoding (C=00, U=01, A=10, G=11) yields the largest depletion (3.4-fold). The encoding-independent $H(3,4)$ result is constant across all 24 bijections and is the primary topology-avoidance test in this work.

S5 Reassignment candidate-universe denominator sensitivity

The hypergeometric and permutation tests of topology avoidance, and the conditional-logit denominator, both depend on a definition of the candidate-move universe $\mathcal{M}(C)$ from the standard code C. Three definitional choices interact: whether stop-codon targets are admitted, whether identity moves ($y = C(x)$) are excluded, and whether the source codon may itself be a stop. The combinations give the variants below; we adopt U1 as primary throughout and report U2 and U4 as sensitivity universes.

Formally, the primary candidate-universe is $\mathcal{M}(C) = \{(x, y) : x \in \mathcal{C}, y \in \mathcal{A}_{20} \cup \{\text{Stop}\}, y \neq C(x)\}$ with $|\mathcal{M}(C)| = 64 \times 20 =$1,280.

Universe	$\lvert\mathcal{M}\rvert$	Definition
U1: 21-label, no identity (primary)	1,280	$y \in \mathcal{A}_{20} \cup \{\text{Stop}\}$, $y \neq C(x)$; each codon has exactly 20 alternative labels
U2: AA-only, no identity	1,219	$y \in \mathcal{A}_{20}$, $y \neq C(x)$; 61 sense codons get 19 AA alternatives, 3 stop codons get 20 AA alternatives
U3: AA, with no-ops	1,280	$y \in \mathcal{A}_{20}$, no identity restriction; 64 candidates are no-ops $y = C(x)$
U4: stop-inclusive with no-ops	1,344	$y \in \mathcal{A}_{20} \cup \{\text{Stop}\}$, no identity restriction; 64 no-ops included

We adopt U1 as the primary universe because it (i) has a uniform alternative count per codon, (ii) cleanly excludes identity moves which contribute no signal, and (iii) admits stop-codon reassignment which is biologically attested. Topology-avoidance results under U2 and U4 are qualitatively identical (depletion remains $p < 10^{-5}$); the per-cell counts differ only by the constant rescaling $K_2 = K_1 -$ (stop-target candidates) for U2. Detailed per-universe numbers are released as part of the `codontopo` repository's `output/` artifacts (the `denominator_sensitivity` block in the topology-avoidance results file).

S6 Conditional logit: IIA assumption and explanatory framing

The conditional-logit framework assumes Independence of Irrelevant Alternatives (IIA): for any two candidate moves m_1 and m_2 in a choice set $\mathcal{N}$, the relative probability $P(m_1)/P(m_2)$ is the same regardless of which other candidates are in $\mathcal{N}$. In the reassignment context, candidate moves are not exchangeable: a move that targets an amino acid already serviced by many codons is plausibly more substitutable with similar moves than the IIA structure allows.

We adopt IIA here because the goal is *explanatory* rather than *predictive*: the test asks whether topology adds explanatory value beyond physicochemical cost (LR test, ΔAICc) within the same candidate set, not whether the model accurately predicts which specific reassignment will occur next. The relative-probability ratios that IIA constrains are not the quantities we report; the LR statistics depend only on whether the observed event occupies a high-likelihood position within the candidate set, which is robust to substitution patterns among non-observed candidates. A mixed logit relaxing IIA would be more appropriate if the goal were prediction; future work could pursue this once a larger event set permits identification of mixed-logit covariance parameters. The posterior-predictive simulation reported in the main text (§3.5; observed topology-breaking rate 0.076 vs simulated mean 0.077; $p = 0.60$) provides a calibration check that the chosen explanatory model reproduces the marginal feature distribution rather than just maximizing in-sample AICc.

S7 Conditional logit: clade-exclusion sensitivity

The conditional-logit framework can confound a single clade's reassignment events with a global topology-avoidance signal if that clade is unusually extensive (or unusually idiosyncratic). To bound this concern, we refit M1–M4 under each of seven clade-exclusion regimes, removing the indicated NCBI translation tables from the event-step list and re-running the full conditional-logit pipeline end-to-end (build candidate sets → enumerate event orderings up to $k!$ for $k \leq 6$ → vectorised maximum-likelihood fit → ΔAICc against M1 and M2). The seven exclusion regimes match those applied to the hypergeometric topology-avoidance test (§S9), which themselves follow the phylogenetic-distribution analysis in Sengupta et al. (2007).

Excluded clade	Tables out	n events	ΔAICc(M1→M3)	ΔAICc(M2→M3)	ΔAICc(M3→M4)
ciliates	6, 10, 15, 27, 28, 29, 30	52	83.7	31.6	1.8
yeast mito	3	60	103.1	87.7	2.2
CUG clade	12, 26	64	116.7	94.9	2.2
metazoan mito	2, 5, 9, 13, 14, 21	40	48.8	88.6	0.4
algal mito	16, 22	63	115.5	85.7	2
protist mito	4, 23	63	100.7	92.4	1.9
hemichordate mito	24, 33	59	92.3	83.5	2.1

Table 2: Per-regime conditional-logit clade-exclusion sensitivity (7 regimes). Each row refits M1–M4 with the indicated NCBI translation tables removed. M3 (physicochemistry + Q_6 topology) remains robustly favored over both M1 (physicochemistry only) and M2 (topology only) in every regime. The minimum ΔAICc(M1→M3) across all seven regimes is 49 the maximum is 117. The conventional Burnham–Anderson reference of ΔAICc>10 was calibrated on linear-regression contexts, so we use it as a reference threshold rather than a formally calibrated cut-off in this conditional-logit setting; the posterior-predictive simulation reported in §3.5 (observed topology-breaking rate 0.076 vs simulated 0.077, $p = 0.60$) is the more directly interpretable calibration check. ΔAICc(M3→M4) values near 2 indicate that adding the heuristic tRNA-distance proxy provides no incremental fit.

S8 Per-table optimality: standard-code-proximity audit

This audit grounds the disaggregation reported in main-text §3.3, where we separated the per-table BH-FDR result into "informative-distance" tables (≥ 3 reassignments from standard) and "near-standard" tables (≤ 2 reassignments).

The methodological concern: a variant code that differs from the standard by only a few reassignments has a per-table block-preserving null distribution dominated by permutations very close to the standard code, simply because few permutations of the variant's block structure yield codes that are far from standard. In that limit, "table X falls in the bottom 5% of permutations preserving X's block structure" partly tests whether X is close to the standard code (which it is, by construction), not whether X is independently optimal.

To address this, for each NCBI translation table we computed two quantities alongside the unconditional per-table quantile: (i) the Hamming distance from the standard code (number of codons with different AA labels) for both the observed variant code and each block-preserving null draw; and (ii) the variant's null quantile **conditional** on null draws that are within ± 2 codons of the variant's distance from standard. If the conditional and unconditional quantiles agree, the per-table optimality is independent of standard-code proximity; if they diverge sharply, the per-table test is largely a proximity test.

Detailed per-table results are released as part of the `codontopo` repository's `output/` artifacts (the `per_table_proximity_audit` block in the coloring-optimality results file). The audit confirms that for variant tables differing from standard by 3 or more codon reassignments, the conditional-on-distance quantile is similar to the unconditional quantile, supporting the interpretation that variant codes preserve error-minimization structure independently of their similarity to the standard code. For variants with ≤2 reassignments, the per-table null contains few sufficiently-distant draws and the test cannot distinguish proximity from independent optimality; we therefore treat those cases as confirmatory of standard-code geometry rather than as evidence for variant-specific optimization.

S9 Topology avoidance: clade-exclusion sensitivity (hypergeometric/permutation)

This section provides the hypergeometric/permutation counterpart to the conditional-logit clade-exclusion analysis in §S7. The two tests probe related but distinct quantities: §S7 tests whether the Δ_{topo} regression coefficient survives clade exclusion; this section tests whether the global landscape-vs-observed depletion ratio survives clade exclusion. Following the phylogenetic-distribution analysis of mitochondrial reassignment events by Sengupta et al. (2007), we iteratively excluded:

- All ciliate reassignments
- All metazoan mitochondrial reassignments
- All CUG-clade yeast reassignments
- All chlorophycean reassignments

In every exclusion, the depletion remains highly significant ($p < 10^{-5}$), confirming that the ≈ 3.4-fold (Q_6) and ≈ 3.1-fold ($H(3,4)$) depletion of topology-breaking changes is a pan-taxonomic pattern, not an artifact of any single lineage. Excluding yeast mitochondrial (table 3) actually strengthens the depletion (rate drops from 21.4% to 8.3%; hypergeometric p from 1.6×10^{-8} to 3.6×10^{-11}), because table 3 contributes 4 of the 6 topology-breaking events. Detailed per-clade counts are released as part of the `codontopo` repository's `output/` artifacts (the phylogenetic-sensitivity results file).

S10 Complete tRNA gene count data

This section documents the empirical foundation for the tRNA enrichment analysis in main-text §3.6. We re-ran tRNAscan-SE 2.0.12 on the 18 NCBI assemblies listed below to produce a uniform tRNA-prediction protocol across the 24 pairings; main-text Table 8 reports the Total column from this analysis, while §S10.1 below resolves Total into its five mutually-exclusive components (Std20 + SeC + Supp + Undet + Pseudo) so the reader can see the per-genome composition.

S10.1 tRNAscan-SE verified organisms

All tRNA gene counts were obtained by running tRNAscan-SE 2.0.12 (Chan & Lowe, 2019) with Infernal 1.1.4 on NCBI genome assemblies. Eukaryotic organisms were scanned in `-E` mode; *Mycoplasma* species in `-B` (bacterial) mode. Total counts are Infernal-confirmed (the more conservative count after the second-pass filter) and match the Total column of main-text Table 8. The five-column breakdown below sums exactly to Total: Std20 (decoding the standard 20 amino acids) + SeC (selenocysteine, anticodon UCA, scanned via the dedicated `TRNAinf-euk-SeC.cm` model) + Supp (possible suppressor tRNAs with CTA/TTA/UCA anticodons) + Undet (predicted tRNAs whose isotype could not be determined) + Pseudo (predicted pseudogenes filtered by the Infernal second-pass isotype validation).

Organism	Tbl	Assembly	Total	Std20	SeC	Supp	Undet	Pseudo	Reassigned AA
T. thermophila	6	GCF_000189635.1	718	672	1	37	4	4	Gln: 54 (15+39)
P. tetraurelia	6	GCF_000165425.1	216	202	1	11	0	2	Gln: 18 (7+11)
O. trifallax	6	GCA_000295675.1	94	83	2	6	2	1	Gln: 8 (2+6)
P. persalinus	6	GCA_001447515.1	262	228	1	15	0	18	Gln: 20 (5+15)
H. grandinella	6	GCA_006369765.1	130	121	1	3	0	5	Gln: 9 (6+3)
E. aediculatus	10	GCA_030463445.1	80	76	1	2	1	0	Cys: 4 (3+1 UCA)
E. amieti	10	GCA_048569255.1	120	103	1	6	1	9	Cys: 8 (4+4 UCA)
E. focardii	10	GCA_001880345.2	62	56	1	3	0	2	Cys: 3 (1+2 UCA)
E. parawoodruffi	10	GCA_021440025.1	149	128	0	3	3	15	Cys: 9 (5+4 UCA)
E. weissei	10	GCA_021440005.1	495	390	0	12	14	79	Cys: 20 (17+3 UCA)
E. woodruffi	10	GCA_027382605.1	83	74	1	3	1	4	Cys: 4 (2+2 UCA)
B. stoltei[§]	15	GCA_965603825.1	169	165	1	0	1	2	Trp: 6
B. nonstop P57	31	GCA_028554745.1	68	65	1	2	0	0	Trp: 2[*]
M. genitalium	4	GCA_000027325.1	36	35	0	1	0	0	Trp: 1[†]
M. pneumoniae	4	GCF_910574535.1	37	36	0	1	0	0	Trp: 1[†]
S. coeruleus	1	GCA_001970955.1	272	265	1	1	3	2	—
I. multifiliis	1	GCF_000220395.1	150	141	1	8	0	0	—
F. salina	1	GCA_022984795.1	89	85	1	1	1	1	—

Table 3: Complete tRNAscan-SE 2.0.12 results (Infernal-confirmed counts). The five tRNA categories sum exactly to Total: **Std20** = standard 20-AA tRNAs; **SeC** = selenocysteine (anticodon UCA); **Supp** = possible suppressor tRNAs (CTA/TTA/UCA anticodons that could read stop codons); **Undet** = predicted tRNAs whose isotype could not be determined; **Pseudo** = predicted pseudogenes filtered by the Infernal second-pass isotype validation. Reassigned-AA counts include standard and suppressor tRNAs counted toward the reassigned amino acid. [§]*Blepharisma stoltei* is indexed as NCBI translation table 15 (Blepharisma Nuclear Code, defined as UAG→Gln) by virtue of its genus assignment, but the strain-specific MAC genome reads UGA→Trp via a dedicated suppressor tRNA-Trp(UCA) (Singh et al., 2023). The analysis here tests Trp-tRNA enrichment on that empirical reading rather than the legacy table-15 nominal code. [*]Anticodon stem shortening (4-bp stem rather than canonical 5-bp). [†]Post-transcriptional modification (a single tRNA-Trp(CCA) reads both UGG and UGA after base modification).

S10.2 MIS (maximal independent set) analysis

To address non-independence from shared control organisms, we constructed a conflict graph where edges connect pairings sharing an organism. The Bron–Kerbosch algorithm with pivoting enumerates all maximal independent sets (MIS). Each MIS represents a set of pairings where no two share an organism and no additional pairing can be added without creating a conflict.

With 24 total pairings, 2 MIS of size 6 were identified. Both are significant at $p < 0.05$ under Stouffer's Z combination of per-pairing Fisher exact p-values:

- **Best-case MIS** ($p = 0.044$): *S. cerevisiae* mito/Thr, *S. obliquus* mito/Leu, *P. tannophilus*/Ala, *P. tetraurelia*/Gln, *E. aediculatus*/Cys, *E. weissei*/Cys
- **Worst-case MIS** ($p = 0.045$): *S. cerevisiae* mito/Thr, *S. obliquus* mito/Leu, *C. albicans*/Ser, *P. tetraurelia*/Gln, *E. aediculatus*/Cys, *E. weissei*/Cys

S10.3 *Saccharomyces cerevisiae* literature-derived control

The *Saccharomyces cerevisiae* tRNA gene counts used in the yeast-mito Thr disconnection pairing come from GtRNAdb (Chan & Lowe, 2016) rather than tRNAscan-SE 2.0.12 in this work. We retain them to keep the well-characterized yeast-mito Thr case in the pairing set and flag the difference in source explicitly.

S11 Complete reassignment database

The reassignment database underpins both the topology-avoidance test (§3.4) and the conditional-logit fits (§3.5). It comprises all codon reassignment events across the 27 NCBI translation tables, relative to the standard code (Table 1). Each event records the codon, source amino acid, target amino acid, and Hamming distance to the nearest codon already encoding the target amino acid in the standard code. De-duplication to unique (codon, target amino acid) pairs yields the event set used in the topology-avoidance analysis. Exact counts (which depend on the most recent NCBI gc.prt revision, currently v4.6 retrieved 2026-04-25) are reported inline in the main text from the same pipeline run that generated this supplement.

The complete database is released as `output/tables/T10_reassignment_db.csv` in the `codontopo` repository.

S12 Synthetic-biology feasibility score (visualization-only)

For Figure 4A of the main text we use a heuristic feasibility score $S(m) \in [0, 1]$ for each candidate single-codon reassignment m from the standard code. The score is **not** used in any inferential test in the manuscript; it is a visualization aid for delineating high- versus low-feasibility regions of the 1,280-move candidate landscape. We report it in detail here for completeness.

The score combines three structural factors:

1. **Filtration preservation** $I_{F(m)} \in \{0, 1\}$: indicator of whether the post-reassignment code retains a uniform 4-bit prefix for any 4-fold-degenerate amino acid (i.e., whether the reassignment preserves the four-fold prefix filtration described in main text §2.2).
2. **Local mismatch smoothness** $f\left(\Delta_{\text{phys}(m)}\right)$: a smooth decay of the move's local Grantham mismatch change, $f\left(\Delta_{\text{phys}}\right) = \exp\left(-\frac{\max(0, \Delta_{\text{phys}})}{d_0}\right)$ with $d_0 = 100$ Grantham units. Negative Δ_{phys} (improving local mismatch) saturates to 1; positive Δ_{phys} decays toward 0.
3. **Accessibility** $g\left(\Delta_{\text{acc}(m)}\right)$: $g(\Delta_{\text{acc}}) = \frac{7-\Delta_{\text{acc}}}{6}$, where $\Delta_{\text{acc}(m)}$ is the minimum Hamming distance from the reassigned codon to any codon currently encoding the target amino acid. Range $[0, 1]$ with full credit at distance 1, zero credit at distance 7.

The composite score is the equal-weighted average:

$$S(m) = \left(\frac{1}{3}\right) I_{F(m)} + \left(\frac{1}{3}\right) f\left(\Delta_{\text{phys}(m)}\right) + \left(\frac{1}{3}\right) g\left(\Delta_{\text{acc}(m)}\right).$$

Filtration-preserving variants ($I_F = 1$) score ≥ 0.8 under this composite; filtration-breaking variants ($I_F = 0$) score ≤ 0.75. The visual gap between the two groups in Figure 4A is therefore mostly driven by I_F. Exact implementation: `src/codon_topo/analysis/synbio_feasibility.py` in the `codontopo` repository.

S13 KRAS–Fano clinical prediction: detailed results

The KRAS–Fano clinical prediction is the most concrete biomedical extrapolation considered for the $\mathrm{GF}(2)^6$ framework, and we tested it directly to delimit scope. The conjecture: XOR ("Fano") relationships in $\mathrm{GF}(2)^6$ predict enrichment of specific amino acids at KRAS G12 co-mutation sites. We tested it against 1,670 KRAS mutations from the MSK-IMPACT dataset (Zehir et al., 2017).

For each of the 6 KRAS G12 variant types (G12D, G12V, G12C, G12A, G12R, G12S), we identified the XOR-predicted amino acid partners and tested for co-mutation enrichment via Fisher's exact test with Bonferroni correction. All 6 tests yielded $p = 1.0$, with odds ratios near 1.0.

This result cleanly separates code-level error-minimization (which operates on the amino acid assignment structure) from mutation-level algebraic predictions (which would require DNA polymerase errors to respect the binary encoding — a biologically implausible mechanism).

S14 Event-level conditional logit model: full details

This section gives the implementation-level detail for the discrete-choice analysis reported in main-text §3.5: the candidate universe, the three feature definitions, the optimisation routine, the order-averaging procedure for unknown event sequences, the fitted raw and normalised coefficients, the likelihood-ratio tests, the predictor-confounding diagnostic, and the per-event observed-move ranks. The complete fitted JSON output is also released as `output/evolutionary_simulation.json` in the public repository.

S14.1 Model specification and candidate universe

The conditional logit model treats each observed reassignment as a discrete choice from the set of all single-codon reassignments available at the current code state. For a code with 64 codons and 21 possible amino acid/stop labels, the candidate set $\mathcal{N}(C)$ contains exactly 1,280 moves (universe U1 in §S5: 64 codons $\times$ 20 alternative labels, excluding identity assignments). All 1,280 candidates are evaluated at each step regardless of biological plausibility; the model's purpose is to test whether the observed moves are statistically distinguishable from uniform sampling given the three feature classes. As a conditional logit, the model implies an independence of irrelevant alternatives (IIA) structure within each choice set; the explanatory–rather–than–predictive framing (§S6) makes IIA tolerable for our purposes.

S14.2 Feature definitions

Local physicochemical mismatch change (Δ_{phys}): For a reassignment of codon c from amino acid a to a', this is the change in the sum of Grantham (1974) distances across all Hamming-1 edges incident to c:

$$\Delta_{\text{phys}} = \sum_{\{c,c'\}:d(c,c')=1} \left[\Delta(a', \text{code}(c')) - \Delta(a, \text{code}(c'))\right]$$

This captures the local impact on error-minimization at the reassigned position. When a neighboring codon is assigned Stop, we set $\Delta(a, \text{Stop})$ equal to the maximum Grantham distance (215), consistent with the stop-penalty convention used in the coloring objective (Methods §2.2 of the main manuscript).

Topology disruption ($\Delta_{\text{topo,Q_6}}$): The total increase in connected components (at $\varepsilon = 1$) summed across all amino acid codon graphs under Q_6 adjacency. Stop codons are excluded from the topology sum; Δ_{topo} counts connected-component changes only for amino acid codon families. A move that splits one amino acid's codon family into two components contributes $+1$; a move that fragments two families contributes $+2$; topology-preserving moves contribute 0. The default $\Delta_{\text{topo,Q_6}}$ uses Q_6 adjacency (Hamming-1 in the default $\mathrm{GF}(2)^6$ encoding $C \to 00$, $U \to 01$, $A \to 10$, $G \to 11$). Because Q_6 adjacency is encoding-dependent (8 of 24 base-to-bit bijections give no Q_6 topology depletion at the candidate-landscape level; §S4), we additionally compute $\Delta_{\text{topo,H(3,4)}}$ using the encoding-independent $H(3,4)$ adjacency (two codons are neighbors iff they differ at exactly one nucleotide position) and refit two model variants: $\text{M2}_{\text{H(3,4)}}$ (topology-only with $\Delta_{\text{topo,H(3,4)}}$) and $\text{M3}_{\text{H(3,4)}}$ (physicochemistry + $\Delta_{\text{topo,H(3,4)}}$). The encoding-robustness comparison $\Delta\text{AICc}(\text{M1} \longrightarrow \text{M3}_{\text{H(3,4)}})$ vs $\Delta\text{AICc}(\text{M1} \longrightarrow \text{M3})$ is reported in the next subsection.

tRNA complexity proxy (Δ_{tRNA}): The Hamming distance from the reassigned codon to the nearest codon already encoding the target amino acid in the current code. This serves as a heuristic for the tRNA repertoire change required to service the reassigned codon, with larger distances implying more novel tRNA machinery needed.

S14.3 Fitted coefficients

Model	Feature	$\hat{\beta}$ (raw)	$\hat{\beta}$ (normalized)
M1	Δ_{phys}	−0.0049	−1.33
M2 (Q_6)	$\Delta_{\text{topo,Q_6}}$	−3.5838	−1.67
M3 (Q_6)	Δ_{phys}	−0.0043	−1.17
M3 (Q_6)	$\Delta_{\text{topo,Q_6}}$	−3.3158	−1.54
M4	Δ_{phys}	−0.0043	−1.17
M4	$\Delta_{\text{topo,Q_6}}$	−2.9855	−1.39
M4	Δ_{tRNA}	−0.2033	−0.21
M2 ($H(3,4)$)	$\Delta_{\text{topo,H(3,4)}}$	−3.6957	−1.79
M3 ($H(3,4)$)	Δ_{phys}	−0.0046	−1.24
M3 ($H(3,4)$)	$\Delta_{\text{topo,H(3,4)}}$	−3.452	−1.67

Table 4: Conditional logit coefficient estimates from the 27-table re-fit. Raw coefficients are on the original feature scale; normalized coefficients are on z-scored features (multiplying raw $\hat{\beta}$ by the global feature standard deviation). All $\hat{\beta}$ values are negative, indicating that observed reassignment histories preferentially populate moves that reduce physicochemical mismatch, avoid topology disruption, and (weakly) prefer target amino acids already serviced by nearby codons. The tRNA proxy coefficient is small and non-significant (LR $= 0.12$, $p = 0.73$). The $H(3,4)$ verification variants ($\text{M2}_{\text{H(3,4)}}$, $\text{M3}_{\text{H(3,4)}}$) replace the encoding-dependent $\Delta_{\text{topo,Q_6}}$ feature with the encoding-independent $\Delta_{\text{topo,H(3,4)}}$. Their fitted ΔAICc values match the main-text encoding-robustness numbers.

S14.4 Likelihood ratio tests

Restricted	Full	LR	df	p
M1 (phys)	M3 (phys+topo, Q_6)	110.4	1	$\ll 10^{-10}$
M2 (topo, Q_6)	M3 (phys+topo, Q_6)	91.2	1	$\ll 10^{-10}$
M3 (phys+topo, Q_6)	M4 (full)	0.1	1	0.73
M1 (phys)	$\text{M3}_{\text{H(3,4)}}$ (phys+topo, $H(3,4)$)	93.4	1	$\ll 10^{-10}$
$\text{M2}_{\text{H(3,4)}}$ (topo, $H(3,4)$)	$\text{M3}_{\text{H(3,4)}}$ (phys+topo, $H(3,4)$)	97.3	1	$\ll 10^{-10}$

Table 5: Likelihood-ratio tests for nested conditional logit models, including both Q_6 topology variants (legacy primary) and $H(3,4)$ topology verification variants. Both topology features (Q_6 added to M1, H(3,4) added to M1) and physicochemistry (added to M2 / $M2_{H(3,4)}$) provide highly significant improvements. The tRNA-complexity proxy does not improve on the phys+topo model.

S14.5 Confounding diagnostic

Across ≈84,000 candidate moves pooled from all choice sets, the Spearman correlation between Δ_{phys} and $\Delta_{\text{topo,Q_6}}$ is $\rho = 0.15$, indicating that the two predictors carry largely independent information. Moves that reduce physicochemical mismatch are only slightly more likely to also preserve topology, and the conditional logit framework accounts for any residual collinearity through simultaneous estimation.

S14.6 Observed move percentile ranks

Under the best model (M3), observed natural reassignments rank at the 89.5th percentile on average (mean rank 134 out of 1,280 candidates). Notable individual events: UGA→Trp ranks consistently above the 98th percentile across all tables where it occurs, indicating that this reassignment is among the most "predictable" given the combined phys+topo score. The one clear outlier is the yeast mitochondrial CUU →Thr reassignment (30th percentile), consistent with translation table 3′s status as the sole marginal exception in the per-table optimality analysis.

S14.7 Order-averaging implementation

For tables with $k > 1$ reassignment events, the temporal ordering is unknown. We marginalized the conditional logit likelihood over all $k!$ orderings, computing $L_{\text{table}} = (1/k!) \sum_{\sigma} \prod_{s} P\big(m^*_{\sigma(s)} \mid \mathcal{N}(C_{\sigma,s})\big)$ using log-sum-exp for numerical stability. For all tables with $k \leq 6$ events (which covers every table in the dataset, including the largest — yeast mitochondrial, $k = 6, 6! = 720$ orderings), we enumerate all orderings exactly. The implementation precomputes feature matrices once per (table, ordering, step) into stacked numpy arrays, after which each Nelder–Mead / L-BFGS-B function evaluation reduces to a small batch of matrix multiplies and `scipy.special.logsumexp` calls; the four nested models are fit concurrently on a single shared bundle via `joblib` threads to avoid memory duplication.

S15 Software and reproducibility

This section provides the metadata a reviewer or replication attempt needs to reproduce every number in the manuscript and supplement bit-for-bit from the public repository. Every figure, table, and inline statistic is rendered by the Typst sources `manuscript.typ` and `supplement.typ` (also in the repository) from the JSON outputs of a single `codon-topo all` invocation, so the manuscript and supplement cannot drift from each other within a single pipeline run.

All analyses were performed using the `codon-topo` Python package (version 0.4.0, commit `0ab163d`). The code is publicly released at https://github.com/biostochastics/codontopo. Dependencies and runtime requirements:

- Python 3.11, NumPy 1.24+, SciPy 1.10+
- R 4.5, ggplot2, ggpubr, viridis, patchwork (for figures)
- tRNAscan-SE 2.0.12 with Infernal 1.1.4
- Random seed: 135325 (all Monte Carlo analyses)

Analyses are fully reproducible via:

```
git clone https://github.com/biostochastics/codontopo.git
cd codontopo
```

```
pip install -e ".[dev]"
codon-topo all --output-dir=./output --seed=135325
python scripts/finalize_manuscript_stats.py
python scripts/generate_tables.py
Rscript src/codon_topo/visualization/R/strengthened_figures.R
```

The complete test suite (432 tests) can be run with `python3.11 -m pytest tests/ -m "not slow"`.

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